

Cost-to-serve and margin analysis

Companion to the cost-model spreadsheet. Real numbers for what each customer actually costs us, what margin remains, and how the math changes at scale.

Why this document exists. Earlier materials cited "\$1,900 variable cost per customer / year, ~80% gross margin." Those numbers were rough. This is the proper model: 78 input variables, three deployment models compared, three customer sizes, scale curves from 10 to 1,000 customers, with each formula traceable to its assumption. **The conclusions are more sober than the earlier figures implied,** and you should know them before any CFO conversation.

Headline findings — with default assumptions

\$1,011 Small variable cost/yr (Model A shared)	\$3,840 Medium variable cost/yr	\$11,543 Large variable cost/yr	~60% Realistic gross margin at scale
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What's true	The number
Multi-tenant shared (Model A) is by far the cheapest	Cost-per-customer ranges \$1K-\$12K/yr depending on size, before fixed-cost amortization.
AI is the single largest cost component	~40-70% of variable cost at Medium/Large customers. Most cost reduction comes from AI optimization, not infra.
Single-tenant isolated (Model C) is 3-8× more expensive	Cost-per-customer \$7.7K-\$20K/yr. Only worth it for enterprise customers paying premium prices.
Gross margin is healthy for Small/Medium, tight for Large	Full price: 78% (S), 64% (M), 48% (L). Heavy AI usage on Large eats margin.
Founding Customer Year 1 is a real acquisition investment	Loss of \$1.3K (S), \$5.8K (M), \$17K (L) per customer in Year 1.
Founding Customer Year 2+ rebuilds to ~50-70% margin	Payback period: 7 months (S), 13 months (M), 19 months (L). 5-year LTV: \$7.7K, \$15.8K, \$26.9K.
Scale economics work as expected	At 100+ customers, fixed costs amortize from \$6,360/customer (at 10 customers) down to ~\$636/customer.

How this differs from prior documents. The business-plan figure of "\$1,900 variable cost / 80% margin" was a Small-customer reference point, applied across all customers. The proper model shows margin **degrades as customer size grows** (more AI usage), not improves. This doesn't break the business — the math still works — but the Medium and Large customers are less profitable than a simple "80% margin SaaS" mental model implies. Update the pitch language accordingly.

How the model is built

Architecture

The model has 6 sheets, formula-driven (300 formulas), with all assumptions on one sheet and everything else derived. Change any input, the entire model recalculates.

Sheet	What it does
0_README	Color conventions, how to use, key conclusions.
1_Assumptions	Every input variable (78 of them). Blue = editable. Sourced.
2_Component_Costs	Per-customer cost build, broken down by line, for three deployment models.
3_Scale_Effects	Cost per customer at 10, 25, 50, 100, 250, 500, 1000 customers.
4_Margin_Analysis	Revenue vs cost, by pricing tier (full, FC Yr 1, FC Yr 2+). LTV + payback.
5_Sensitivities	What breaks the model: AI 2x, egress spike, support load, etc.

Customer-size definitions

Size	Profile	Sites	Users	Audits/yr
Small	Tier 4 SME / nutraceutical, single facility	1	5	12
Medium	Tier 3 CDMO / Tier 2 mid-pharma, multi-site	3	20	30
Large	Tier 2 large + multi-site, heavy AI usage	5	50	60

Deployment-model definitions

Model	What it is	Trade-off
A — Multi-tenant shared	One AWS cluster, one MongoDB cluster, all customers isolated by tenant ID	Cheapest. The default for SMB SaaS. Most customers prefer it once they understand the security model.
B — Hybrid	Small/Medium on shared; Large on dedicated infra	The realistic mid-stage state — gives enterprise customers what they ask for without paying isolation cost for everyone.
C — Single-tenant isolated	Every customer gets their own MongoDB cluster, compute, storage	Most expensive. Only worth it for: data-residency mandates, on-prem-equivalent requirements, strict CSV/validation rules. 3-8x the cost.

Pricing assumptions used

Tier	ACV (full)	Impl fee	Notes
Small	\$4,500	\$2,500	Per-site (1) + 5 users + AI credits
Medium	\$10,800	\$5,000	Per-site (3) + 20 users + AI credits. ~₹9L — the deck's headline ACV.
Large	\$22,000	\$12,000	Per-site (5) + 50 users + heavier AI usage

Cost-line breakdown — Model A multi-tenant shared

Medium customer cost breakdown (\$3,840/yr total)

Cost line	Annual (\$)	% of total	What drives it
AI inference (frontier APIs, weighted mix)	~\$1,035	27%	250M input + 60M output tokens/yr at blended \$2.10/\$8.50 per M
Customer success & support	~\$1,800	47%	72 hrs CSM/yr + 20% engineering escalation
MongoDB Atlas (prorated share)	~\$591	15%	25GB DB on shared M30 cluster (200GB capacity)
Compute (prorated)	~\$153	4%	2 t3.medium + 2 m5.large workers prorated by usage
S3 storage + backup + egress	~\$70	2%	150GB documents + backups + 50GB egress/yr
OCR / document parsing	~\$18	0%	12K pages/yr at \$0.0015/page
Observability (CW + Datadog share)	~\$150	4%	Logs + metrics per tenant
Total variable cost / Medium customer / yr	~\$3,840	100%	

The most important reading of this table: support (47%) and AI (27%) together are 74% of variable cost. Cloud infra and storage are noise. The two real cost-control levers are: (1) **reducing support load per customer** — better self-service, in-product help, fewer escalations — and (2) **moving AI to self-hosted open-source models in Year 2+** — can cut the AI line by 60-70%. Optimizing AWS is the wrong fight.

How cost grows with customer size

Cost line	Small	Medium	Large	Scaling factor
AI inference	\$190	\$1,035	\$3,930	Linear with usage; biggest driver of size differential
Support & CS	\$598	\$1,800	\$4,500	Linear with users/audits/sites
MongoDB	\$118	\$591	\$1,892	Linear with DB size
Compute	~\$45	~\$153	~\$385	Prorated by usage
Other (storage, OCR, observability, egress)	~\$60	~\$260	~\$835	Various
Total / customer / year	~\$1,011	~\$3,840	~\$11,543	

Reading note. The Large-customer line (\$11,543/yr cost) against a \$22,000 ACV looks tight (48% margin). It is. Two reasons: (a) the Large profile assumes very heavy AI usage (900M input tokens/yr — aggressive); (b) implementation hours scale faster than revenue at large sizes. Both can be optimized: AI usage by routing more traffic to cheaper models, implementation by templates/partner-led delivery. The model lets you stress-test these.

Revenue, cost, and margin at each pricing tier

Tier 1 — Full price (Year 2+ for non-Founding Customers)

Metric	Small	Medium	Large
Annual revenue (ACV)	\$4,500	\$10,800	\$22,000
Variable cost to serve (Model A)	\$1,011	\$3,840	\$11,543
Gross profit (before fixed)	\$3,489	\$6,960	\$10,457
Gross margin %	77.5%	64.4%	47.5%

Tier 2 — Founding Customer Year 1 (free software + impl fee)

Metric	Small	Medium	Large
Software subscription revenue	\$0	\$0	\$0
Implementation fee revenue (one-time)	\$2,500	\$5,000	\$12,000
Total Year 1 revenue	\$2,500	\$5,000	\$12,000
Variable cost to serve (Year 1)	\$1,011	\$3,840	\$11,543
Implementation cost (one-time, eng time)	\$2,800	\$7,000	\$17,500
Total Year 1 cost	\$3,811	\$10,840	\$29,043
Year 1 gross profit / (loss)	(\$1,311)	(\$5,840)	(\$17,043)

Tier 3 — Founding Customer Year 2+ (50% software + cost-of-serve passthrough)

Metric	Small	Medium	Large
Software subscription (50% of full)	\$2,250	\$5,400	\$11,000
Infrastructure passthrough revenue	\$1,011	\$3,840	\$11,543
Total annual revenue (Year 2+)	\$3,261	\$9,240	\$22,543
Variable cost to serve	\$1,011	\$3,840	\$11,543
Gross profit (Year 2+)	\$2,250	\$5,400	\$11,000
Gross margin %	69%	58%	49%

5-year customer lifetime value (Founding Customer path): Small \$7.7K, Medium \$15.8K, Large \$27.0K. Versus full-price (\$17K / \$35K / \$52K). The Founding Customer offer costs ~50-55% of LTV in exchange for early references. This is defensible for the first 10 customers; beyond that, charge full price.

Cost per customer at scale

As you add customers, fixed shared platform costs (\$63,600/year — observability, security, marketing tools, ops, legal) spread thinner. Variable cost-per-customer stays constant.

Total cost / customer / year, by deployment model and scale

Customer count	10	25	50	100	250	500	1000
Model A — Multi-tenant shared	\$9,555	\$5,739	\$4,467	\$3,831	\$3,450	\$3,323	\$3,259
Model B — Hybrid	\$10,168	\$6,352	\$5,080	\$4,444	\$4,062	\$3,935	\$3,871
Model C — Single-tenant isolated	\$16,710	\$12,894	\$11,622	\$10,986	\$10,604	\$10,477	\$10,413

What this means in plain language.
At **10 customers** (your first 18 months), Model A total cost per customer is ~\$9,555/yr — gross margin against a ~\$9,000-10,000 ACV is essentially zero. **Early-stage SaaS is supposed to look like this.** Fixed costs dominate; you're investing in the platform.

At **50 customers** (Series A milestone), cost drops to ~\$4,467/customer — gross margin against the same ACV is ~55%. **This is when the business starts making sense.**

At **100+ customers** (post-Series A), cost approaches \$3,800/customer — gross margin ~60%. **This is the steady-state SaaS profile.**

At **1,000 customers**, cost is \$3,259 — the variable-only floor (\$3,195) plus negligible fixed allocation. **Beyond ~200 customers, scale doesn't help much — you've already amortized everything.**

Customer mix assumption

The scale model assumes a steady customer mix: **50% Small, 40% Medium, 10% Large**. If your actual mix skews toward Large (premium customers), variable cost per customer increases significantly. The spreadsheet lets you change the mix and see the impact.

Stress tests for the Medium customer

Each row shifts one assumption. Shows variable cost for Medium customer (base: \$3,840/yr) and the delta. This is the analysis a sharp CFO will ask for.

Scenario	Variable cost	Δ vs base	What it means
BASE (Medium, Model A, planned usage)	\$3,840	—	Default
AI tokens 2× (heavy AI adoption)	\$4,875	+\$1,035 (+27%)	Most common upside risk — customers love AI features and use them more than planned
AI tokens 0.5× (light adoption)	\$3,323	-\$518 (-13%)	Customer underuses AI features
Cloud egress 3× (heavy downloads)	\$3,849	+\$9 (+0%)	Egress is negligible, surprisingly. Not a real risk.
Support load 2× (high-touch customer)	\$5,634	+\$1,794 (+47%)	Bad-fit customer needing hand-holding. Avoid via better PoC qualification.
Database 3× (data hoarder)	\$4,896	+\$1,056 (+27%)	Customer retains everything, never archives. Manageable with policy.
Self-hosted AI in Year 2+	\$2,304	-\$1,536 (-40%)	Biggest cost lever. Self-hosting tuned Llama-3 on owned GPUs cuts AI cost ~70%.
Reserved-instance pricing (40% off compute)	\$3,264	-\$576 (-15%)	Available once usage stable. Standard SaaS move.
Cloud egress 10× (mass export event)	\$3,881	+\$41 (+1%)	Still negligible. Egress paranoia is overblown.
Customer churns at month 6	\$1,920	-\$1,920 (-50%)	Half the cost (less usage) but you lose CAC investment.

The three risks worth taking seriously

- Support load > 2× assumed** — the biggest unforeseen cost. Mitigation: tight PoC qualification, written success criteria before commitment, in-product help reducing tickets.
- AI usage spike** — can erode margin fast at Medium/Large. Mitigation: per-customer AI usage caps in product, transparent overage billing, fast track to self-hosting in Year 2.
- Heavy Large-customer skew** — Large customers are less profitable than Small/Medium under current pricing. Either bump Large ACV by 30-40% or optimize their AI cost structure.

What's NOT a risk despite intuition: AWS infra cost. Egress, storage, compute together are <10% of total. Optimizing AWS is a distraction; the leverage is in AI and support.

What this means for everything else

How the prior documents need updating

Document	Stated number	Reality per model
Business plan	"~\$1,900 variable cost/customer"	True only for Small. Medium \$3,840, Large \$11,543. Update to: "\$1-12K depending on size."
Business plan	"~80% gross margin"	True at Small full-price. Medium 64%, Large 48% — lower than implied. Update to: "60-78% gross margin depending on size."
Management deck	"4.5x ROI, 3-4 month payback (customer)"	Customer-side savings (₹38L vs ₹9L paid) numbers hold. The 4.5x is customer ROI, not your margin.
Management deck	"Founding Customer Year 1: implementation profit"	Wrong. Year 1 is a loss of \$1.3-17K per customer once cost-of-serve is included. Frame it correctly: "We absorb cost-to-serve as part of the program."
Pitch deck	"~\$9,500 ACV at 80% margin"	Specifically for Medium customers, the right numbers are \$10,800 ACV at 64% margin. Update.

The honest pitch language going forward

"At our planned customer mix and pricing, gross margin lands between 60-78% depending on customer size — healthy for SaaS, with room to grow toward 75%+ blended as we shift AI workloads to self-hosted models in Year 2+. The Founding Customer offer absorbs ~\$5K/customer in Year 1 as acquisition cost; Year 2+ rebuilds to ~55-65% gross margin. Payback per Founding Customer is 13 months on average."

What to do operationally

- Treat AWS Activate / startup credits as a Year-1 essential.** \$5-25K in free AWS credit eliminates most variable infra cost in your first 12 months. Apply immediately if you haven't.
- Pursue Anthropic Startup Program and OpenAI startup credits.** Can reduce AI line by 50-80% in Year 1.
- Plan self-hosted AI for Month 12-18.** This is the single largest cost lever in the model.
- Set hard internal AI usage caps per customer at signing.** Document them. Charge for overage. Don't be surprised by usage.
- Limit Founding Customer cohort strictly to 10.** Each costs ~\$5K Year 1 acquisition; at 10 that's ~\$50K of investment. That's a meaningful number for your burn rate.
- Track real bills against this model monthly.** Replace assumed numbers with actual numbers as they come in. The model is built to be updated.

One thing the model doesn't capture: support quality risk. The assumption of 72 hrs/Medium customer/yr in support is industry-typical for well-built SaaS. If your platform requires more hand-holding — due to bugs, gaps, or complexity — support cost could 2-3x, killing margin entirely. The model has a sensitivity for this; the cure is product quality, not pricing.

Closing thought. Real unit economics for early-stage AI-native SaaS look like this: tight Year 1, decent Year 2, healthy Year 3+, dependent on AI cost optimization to land at the "great SaaS" margin range of 75-80%. The path is real but it requires execution. The model lets you watch the path in real-time as your real numbers come in.